

VaxiFlow

Risk Management Plan

Last-mile Vaccine Cold Chain Monitoring and Alert System

Three-Month Pilot • Northern Delivery Route • Four Rural Clinics

Document item	Details
Project	VaxiFlow – Last-mile Vaccine Cold Chain Monitoring and Alert System
Document	Risk Management Plan
Version	1.0 – Working Project Document
Project Sponsor	Fatima Al-Sayed, Cold Chain Program Manager
Project Manager	Pending formal appointment
Prepared for	VaxiFlow / Group 6
Planning basis	Project Charter, Business Case, Feature Scope, Presentation Work, and project schedule
Status	Prepared before implementation; risks are based on planned scope, constraints, assumptions, and pilot conditions

Important note

This risk plan is an anticipatory project-management document. The VaxiFlow system has not yet been fully implemented, so risks are assessed from the planned requirements, proposed architecture, pilot environment, schedule, stakeholders, and documented constraints. Likelihood and impact ratings in this document are proposed planning ratings and should be reviewed by the team as the project progresses.

1. Introduction

1.1 Purpose of the Risk Management Plan

The purpose of this Risk Management Plan is to provide a structured approach for identifying, assessing, responding to, monitoring, and controlling risks that may affect the VaxiFlow project. The project proposes a three-month pilot of an offline-capable vaccine cold chain monitoring and alert system on a northern delivery route serving four rural clinics. The current process depends on paper strip-chart recorders and manual checking, and the project is intended to improve the early detection and handling of temperature excursions.

Risk management is particularly important for VaxiFlow because the project combines software, temperature-sensing hardware, offline operation, field delivery activities, clinical governance, data management, and multiple stakeholders. A failure in any one of these areas could affect the pilot schedule, data quality, user adoption, or the ability to detect and respond to a temperature problem.

1.2 Project Context

The project documentation identifies a recent cold-chain incident in which vaccines went outside the safe temperature range and several doses were administered before the problem was discovered. The current paper-based process also makes it difficult to determine when an excursion occurred, how long it lasted, and which cold boxes or vaccine types were affected. The proposed solution therefore focuses on continuous monitoring, local offline alerts, digital records, vaccine-specific rules, and later synchronisation.

Source basis

The Business Case states that the pilot is intended to operate without relying on continuous internet connectivity and to keep reliable digital shipment records. The Project Charter defines continuous temperature monitoring, offline local alerts, vaccine-specific rules, shipment/excursion records, offline storage with later synchronisation, reporting/audit records, and role-based access as high-level requirements. These requirements form the main basis of this risk plan.

2. Risk Management Objectives

The risk management process for VaxiFlow aims to:

- Protect the primary patient-safety objective by reducing the possibility that an undetected temperature excursion reaches patients.
- Identify technical, operational, clinical, regulatory, data, schedule, stakeholder, and scope risks before they become project issues.
- Prioritise risks so that the team can focus limited pilot time and resources on risks with the greatest potential effect.
- Define practical preventive actions and contingency actions for high-priority risks.
- Assign a responsible owner to each major risk so that monitoring and response are not left unassigned.
- Create clear escalation paths for risks involving clinical protocols, vaccine rules, budget, regulation, ministry relationships, or major scope changes.
- Protect the three-month pilot schedule by identifying risks that could delay requirements, prototype development, testing, deployment, or evaluation.

- Maintain a living risk register that can be updated as assumptions are confirmed and implementation begins.

2.1 Risk Management Principles

- Prevention first: reduce the probability of a risk where practical before the field pilot begins.
- Safety first: patient safety and correct clinical handling take priority over convenience or schedule pressure.
- Offline-first thinking: rural connectivity limitations are treated as a design condition rather than as an unexpected exception.
- Clear ownership: every material risk should have a named owner and an agreed response.
- Evidence-based decisions: temperature records, test results, incident records, device reliability data, and user feedback should be used when reviewing risks.
- Controlled change: new features or requirements should be assessed for their effect on scope, schedule, budget, and risk before being accepted.
- Escalation of authority: the Project Manager handles routine project matters but major scope, budget, clinical, vaccine-rule, and ministry-related decisions are escalated to the appropriate authority.

3. Risk Categories

The following categories are used to organise the VaxiFlow risk register.

Category	Description	VaxiFlow examples
Technical / System	Risks related to software, sensors, hardware, data storage, synchronisation, or system behaviour.	Device failure, missed excursion, false alert, offline storage/sync failure.
Connectivity	Risks caused by weak or unavailable internet/mobile coverage.	Delayed synchronisation or remote visibility.
Power / Hardware Operations	Risks related to battery capacity, charging, calibration, and device readiness.	Unreliable clinic power, insufficient battery.
Clinical / Patient Safety	Risks that could affect vaccine handling or the response to an excursion.	Incorrect vaccine rules, unclear incident response, missed excursions.
Regulatory / Compliance	Risks related to certification, governance, retention, or required approvals.	Unknown device certification requirements.
Data / Security	Risks related to ownership, access, retention, auditability, and sensitive incident information.	Unclear data governance.
People / Adoption	Risks related to drivers, nurses, depot staff, training, workload, or acceptance.	Driver resistance, low nurse acceptance.
Stakeholder / Coordination	Risks caused by multiple authorities or stakeholder dependencies.	Cross-district coordination and approvals.
Schedule / Project Control	Risks that threaten the three-month pilot or sequencing of work.	Scope creep, approval delays, PM appointment delay.
Financial	Risks related to the unconfirmed pilot budget and cost assumptions.	Budget confirmation delay or cost changes.

4. Risk Assessment Methodology

Each identified risk is assessed using two dimensions: Likelihood and Impact. The proposed scoring system uses a 1–5 scale. The resulting Risk Score is calculated as Likelihood × Impact.

Rating	Likelihood	Interpretation
1 – Rare	Very unlikely	Not expected under normal pilot conditions, but possible.
2 – Unlikely	Unlikely	Could occur, but there are reasonable controls in place.
3 – Possible	Possible	Could reasonably occur during the pilot.
4 – Likely	Likely	Expected to occur or has strong contributing conditions.
5 – Almost Certain	Very likely	Highly likely to occur unless actively controlled.

Rating	Impact	Interpretation
1 – Minor	Limited effect	Small inconvenience; little or no effect on key deliverables.
2 – Moderate	Manageable effect	May require additional work but is unlikely to threaten a key objective.
3 – Significant	Noticeable effect	Could affect a deliverable, user acceptance, or schedule.
4 – Major	Serious effect	Could materially delay the pilot or compromise an important requirement.
5 – Severe	Critical effect	Could threaten patient safety, cause a major pilot failure, or prevent deployment.

4.1 Proposed Risk Priority Thresholds

Score	Priority	Planning response
1–4	Low	Monitor during normal project reviews; action when conditions change.
5–9	Medium	Assign an owner and preventive action; review regularly.
10–14	High	Active mitigation required; track at project-control reviews.
15–25	Critical	Immediate attention from the responsible lead; escalate where required and prepare contingency action.

These thresholds are proposed for project planning and are not presented as a formally approved organisational risk standard. The team should confirm the scoring convention with the course/project requirements if a prescribed method is provided.

5. Risk Register

The following register is derived primarily from the risks, constraints, assumptions, and project boundaries documented in the Business Case and Project Charter. Additional project-control risks are explicitly marked as planning risks derived from those documents rather than as risks directly listed in the original risk table.

ID	Risk	Category	Cause / Condition	Potential impact	L	I	Score	Priority	Response	Preventive / Response Actions	When to monitor	Owner	Trigger / Warning
R01	Loss of internet/mobile connectivity	Connectivity	Rural route has poor or no internet/mobile coverage; synchronisation may be delayed.	Delayed data synchronisation and reduced remote visibility during delivery.	5	3	15	Critical	Reduce	Design local monitoring and alerts to operate offline; store records locally; synchronise when connectivity returns.	Before design sign-off and during offline testing	Technical / Backend Lead	Connectivity unavailable during a delivery or sync queue remains pending.
R02	Unreliable power supply	Hardware / Operations	Two of the four pilot clinics have unreliable power.	Devices may not be charged or may lose power; monitoring could stop during a delivery run.	4	4	16	Critical	Reduce	Charge/check devices at depot; confirm battery capacity for full journey; define charging and maintenance routine; retain paper backup.	Before every delivery; review during pilot	Northern Depot Manager / Technical Lead	Low battery, missed charging check, or repeated power interruption.
R03	Device failure during transport	Technical / Hardware	Monitoring depends on cold-box devices and sensors operating throughout delivery.	Temperature history may be incomplete or unavailable, making shipment assessment difficult.	3	5	15	Critical	Reduce / Contingency	Pre-deployment hardware testing, calibration and spare units; retain paper logs; record device failures as pilot incidents.	Testing phase and every pilot run	Technical Coordination Lead	Sensor stops recording, device reboots unexpectedly, damaged unit, or missing data.
R04	False temperature alerts	Technical / Clinical	Incorrect thresholds, sensor behaviour, or alert logic could trigger unnecessary warnings.	Staff may lose trust or unnecessarily hold/discard usable vaccines.	3	4	12	High	Reduce	Use approved vaccine-specific rules; simulate alert conditions; validate threshold logic; track false-alert frequency.	During system testing and weekly pilot review	Testing, Integration & Quality Lead / Regional Pharmacist	Alert occurs when test conditions are within the approved range.

R05	Missed temperature excursion	Clinical / Patient Safety	Monitoring or alert logic may fail to detect a real excursion.	Potentially compromised vaccines could reach patients; key patient-safety objective may fail.	2	5	10	High	Reduce / Contingency	End-to-end excursion simulation; test sensor accuracy, logging and local alert path; treat missed excursions as a critical pilot failure measure; retain paper backup.	Before field launch and continuously during pilot	Testing, Integration & Quality Lead	Test excursion not detected, missing readings, or alert path fails.
R06	Driver resistance	People / Adoption	Drivers may interpret monitoring as personal surveillance or may resist the system.	Reduced use, interference, or unreliable operational response during transport.	3	3	9	Medium	Reduce	Explain purpose and recorded information; involve drivers in testing; use event-based location tracking rather than continuous GPS if location information is used.	User testing and throughout pilot	Logistics Coordinator / Depot Manager	Drivers express concerns, bypass functions, or fail to respond to alerts.
R07	Low nurse acceptance	People / Adoption	Nurses are busy with patients and may avoid a process that adds work.	System may be bypassed or shipment status may not be checked consistently.	3	3	9	Medium	Reduce	Keep nurse-facing process simple; use clear status indicators; avoid unnecessary forms/screens; include nurses in UAT and collect feedback.	UAT and field pilot	UI/UX Lead / Logistics Coordinator	Repeated complaints, skipped status checks, or workarounds observed.
R08	Unclear response after an alert	Clinical / Operations	Detecting an excursion does not automatically define who must act.	A temperature problem could be detected but response could be delayed or inconsistent.	3	5	15	Critical	Reduce	Approve an escalation and incident-response protocol before field deployment; define depot, driver, clinic, pharmacist and medical responsibilities.	Before field pilot	Medical Director / Project Sponsor	Alert occurs but users are unsure who should be contacted or what action to take.

R09	Outdated or incorrect vaccine temperature rules	Clinical / Governance	Different vaccines require approved temperature limits/exposure rules; incorrect rules could be used.	Unsafe or unnecessary decisions; false alerts or missed excursions.	2	5	10	High	Avoid / Reduce	Regional Pharmacist approves rules; restrict changes to authorised medical/pharmacy staff; keep audit trail; show rule version/status; review rule updates before use.	Before configuration and whenever rules change	Regional Pharmacist	Rule update received, rule version expires, or unauthorised change detected.
R10	Data governance issues	Data / Compliance	Ownership, access, retention, and possible ministry handover are not yet fully confirmed.	Legal/operational problems, inappropriate access, or unclear retention obligations.	2	4	8	Medium	Reduce	Prepare data governance agreement; define ownership, access roles, retention and handover expectations with Legal, medical and ministry stakeholders.	Before pilot launch	Legal Team / Project Manager	Unresolved data owner, retention requirement, or access dispute.
R11	Unknown device certification requirements	Regulatory	Certification/regulatory requirements for monitoring devices are not yet confirmed.	Deployment could be delayed or a device could be unsuitable for the pilot.	3	4	12	High	Avoid / Reduce	Confirm certification requirements early; involve technical/compliance reviewers before procurement and deployment.	Weeks 1–2	Project Sponsor / Technical Coordination Lead	Certification requirement not discovered after device selection.
R12	Cross-district coordination delay	Stakeholder / Coordination	The northern route covers clinics under two local health authorities.	Approvals or operational coordination may delay deployment or create inconsistent procedures.	3	3	9	Medium	Reduce	Involve both authorities early; document required approvals; maintain a single coordination checklist.	Requirements/approval phase	Logistics Coordinator	Approval remains pending or authorities provide conflicting requirements.

R13	Scope creep	Schedule / Scope	The system has many possible features, while the pilot has a limited three-month duration.	Extra functions could delay core monitoring, alerts, records, testing and reporting.	4	4	16	Critical	Avoid / Reduce	Maintain the pilot boundary; prioritise essential monitoring, local alerts, digital records and basic reporting; assess every requested change for schedule/benefit impact.	At every scope review	Risk, Schedule & Project Control Lead / Project Sponsor	New feature request threatens baseline schedule or core MVP work.
R14	Budget confirmation or cost change	Financial / Schedule	The budget is still to be confirmed and current figures are planning estimates.	Procurement or development could be delayed; planned scope may need adjustment.	3	4	12	High	Reduce	Confirm final budget early; separate essential pilot costs from optional items; maintain contingency; escalate major cost changes.	Weeks 1–2 and at change requests	Project Sponsor / Project Manager	Finance does not approve the estimated budget or supplier quotes exceed assumptions.
R15	Project Manager appointment delay	Governance / Schedule	Project Charter states that the Project Manager is pending formal appointment before commencement.	Day-to-day coordination, escalation and schedule control may be unclear.	3	4	12	High	Reduce	Complete formal appointment before project commencement; temporarily document interim coordination responsibilities if required by the project authority.	Before project commencement	Project Sponsor	No formally appointed PM when implementation activities are due to start.

6. Probability–Impact Matrix

The matrix below visualises the proposed ratings. The score is calculated as Likelihood × Impact. Because patient-safety impacts can be severe even when likelihood is low, risks such as missed excursions remain high priority even with a lower likelihood rating.

	Impact 1	Impact 2	Impact 3	Impact 4	Impact 5
Likelihood 5	5 (M)	10 (H)	15 (C)	20 (C)	25 (C)
Likelihood 4	4 (L)	8 (M)	12 (H)	16 (C)	20 (C)

Likelihood 3	3 (L)	6 (M)	9 (M)	12 (H)	15 (C)
Likelihood 2	2 (L)	4 (L)	6 (M)	8 (M)	10 (H)
Likelihood 1	1 (L)	2 (L)	3 (L)	4 (L)	5 (M)

Legend: L = Low, M = Medium, H = High, C = Critical.

6.1 Priority Risks Requiring Active Attention

- R01 Connectivity: score 15. Connectivity loss is an expected condition of the rural route, so the system must remain useful without a network connection.
- R02 Unreliable power: score 16. Two pilot clinics have unreliable power, making battery readiness and depot charging controls important.
- R03 Device failure: score 15. A hardware failure could remove the temperature evidence needed to assess a shipment safely.
- R08 Unclear response after an alert: score 15. Detection without an agreed response process could still allow delays in handling an excursion.
- R13 Scope creep: score 16. The three-month pilot creates a strong schedule constraint; additional features could reduce time available for the essential functions.

R05 Missed temperature excursion has a proposed score of 10 rather than 15 or higher because the documented project response emphasises thorough pre-launch testing. However, its impact is severe, so it remains a high-priority clinical risk and a critical pilot-failure measure.

7. Risk Response Strategy

7.1 Avoidance

Avoidance will be used when a risk can be removed by changing the project boundary or preventing an uncertain condition from entering the pilot. The clearest example is scope creep: the project should avoid introducing predictive route optimisation, continuous GPS tracking, national health-system integration, advanced analytics, or complex dashboards during the pilot because these are explicitly outside the documented scope.

7.2 Reduction / Mitigation

Mitigation reduces the likelihood or impact of a risk without removing the underlying condition. This is the main response strategy for VaxiFlow. Examples include offline operation for connectivity risk, depot charging checks for power risk, hardware testing and spares for device failure, vaccine-specific rules for false alerts, and user testing for adoption risks.

7.3 Contingency

Contingency actions are used if the risk actually occurs. During the pilot, paper temperature logs are retained as a temporary backup. A device failure should be recorded as an incident, and a failed or uncertain shipment should follow the approved incident-response process rather than being automatically cleared by the system.

7.4 Escalation

Risks involving major scope changes, budget, clinical protocols, vaccine temperature rules, or ministry-related decisions should be escalated. The Project Charter states that the Project Manager can manage routine project activities

but does not have independent authority to approve major changes to scope, budget, clinical protocols, vaccine temperature rules, or ministry decisions.

8. Detailed Risk Response Plans

8.1 Connectivity Failure

The pilot route includes areas with poor or no internet/mobile coverage. Connectivity loss is therefore not simply an exceptional technical failure; it is a normal operating condition that the proposed system must handle.

Response actions:

- Local temperature monitoring must continue without internet access.
- Local alerts must be generated without waiting for a cloud connection.
- Temperature and shipment data must be stored locally.
- A delayed synchronisation queue should transfer records when connectivity returns.
- Testing should include extended offline periods, not only brief network interruptions.

Monitoring focus: The team should monitor pending synchronisation queues, missing records, and any difference between local records and central records after reconnection.

8.2 Power and Battery Risk

The Project Charter identifies unreliable power at two of the four pilot clinics and requires sufficient battery capacity for a complete delivery journey.

Response actions:

- Charge devices at the depot before each journey.
- Record a basic readiness check before dispatch.
- Confirm battery capacity under realistic operating conditions.
- Maintain spare units where available.
- Keep paper logs as a temporary backup during the pilot.

Monitoring focus: Low-battery warnings, failed charging checks, or devices returning with insufficient battery should trigger review of the charging and maintenance process.

8.3 Device Failure

Because temperature monitoring depends on hardware, sensor or device failure can create a gap in evidence.

Response actions:

- Perform hardware-software integration tests before deployment.
- Calibrate devices according to the confirmed technical requirements.
- Use spare units where available.
- Record failures as incidents for evaluation.
- Retain the temporary paper backup until the pilot demonstrates reliable operation.

Monitoring focus: The evaluation should track failure frequency, failure type, affected shipment, recovery time, and whether the failure caused a loss of temperature evidence.

8.4 False or Missed Alerts

The system must detect excursions accurately. Both false alerts and missed alerts are important: false alerts can reduce trust and cause unnecessary holds, while missed alerts can undermine the patient-safety objective.

Response actions:

- Use approved vaccine-specific rules.
- Test normal, boundary, and excursion conditions.
- Test both short and sustained excursions according to the approved requirements.
- Verify that the sensor, rule engine, alert trigger, status indicator, and record creation work together.
- Track false alerts and missed excursions separately during pilot evaluation.

Monitoring focus: Any missed excursion should be escalated immediately as a critical pilot failure measure rather than treated as a routine usability issue.

8.5 Unclear Alert Response

The project documentation identifies the possibility that an alert could be detected without a clear understanding of who should act.

Response actions:

- Approve an incident-response and escalation protocol before field use.
- Define responsibilities for driver, clinic nurse, depot, pharmacist, and medical authority.
- Make the system display a clear hold/seek-advice state rather than making an automatic clinical decision.
- Use training and quick-reference guidance to reinforce the response process.

Monitoring focus: During UAT, test whether users can identify the correct next step after an alert. Any uncertainty should be corrected before deployment.

8.6 Vaccine Rule Governance

Temperature limits and exposure rules are clinically important and must not become uncontrolled configuration values.

Response actions:

- Regional Pharmacist approval before field use.
- Restrict rule changes to authorised medical/pharmacy staff.
- Maintain an audit record of changes.
- Show rule/version status where appropriate.
- Review changes before they are applied to pilot devices.

Monitoring focus: Any change in vaccine guidance, rule version, or authorisation status should trigger a configuration review.

8.7 User Acceptance

Drivers and nurses are frontline users. The documents note that drivers may interpret monitoring as surveillance and nurses may bypass systems that add work.

Response actions:

- Explain the purpose of monitoring and what information is recorded.

- Involve drivers and nurses in user testing.
- Keep nurse-facing status information simple.
- Avoid unnecessary forms and complex screens.
- Collect user feedback during testing and the field pilot.

Monitoring focus: Repeated bypass behaviour, confusion, complaints, or workarounds should be recorded and reviewed as adoption risks.

8.8 Regulatory and Data Governance

The project documents state that device certification and data-retention requirements remain to be confirmed.

Response actions:

- Confirm certification requirements early in the project.
- Consult the Legal Team on ownership, access, retention, and sensitive incident records.
- Document required approvals before field deployment.
- Do not treat regulatory confirmation as a post-deployment task.

Monitoring focus: An unresolved certification or governance question near the deployment decision should be escalated before field use.

8.9 Scope and Schedule Control

The project is deliberately limited to a three-month pilot. The documentation also identifies advanced features that are outside scope.

Response actions:

- Maintain a baseline list of pilot functions.
- Evaluate new requests against patient-safety objectives and pilot success criteria.
- Defer predictive routing, continuous GPS, national-system integration, advanced analytics, complex dashboards, and full rollout.
- Escalate major scope changes to the Project Sponsor.

Monitoring focus: A feature request should be flagged when it requires new hardware, changes the architecture, adds significant testing, or threatens the planned pilot timeline.

9. Risk Ownership and Escalation

Risk ownership follows the roles documented in the project materials. Because the Project Manager is still pending formal appointment, ownership should be confirmed when the project governance structure is finalised.

Role / Stakeholder	Primary risk responsibilities
Project Sponsor – Fatima Al-Sayed	Major scope, budget, stakeholder, ministry and escalation decisions; overall pilot accountability.
Project Manager – pending appointment	Maintain risk register, coordinate mitigation actions, track schedule-related risks, escalate major risks, and report status to sponsor.
Risk, Schedule & Project Control Lead	Maintain risk register support, monitor scope/schedule risks, prepare risk review information.

Project Manager / Technical Coordination Lead	Coordinate technical risks involving architecture, hardware, offline operation and implementation dependencies.
Testing, Integration & Quality Lead	Own test-related controls for device behaviour, missed/false alerts, offline persistence and integration.
Northern Depot Manager	Device readiness, charging, shipment preparation, operational checks and driver support.
Logistics Coordinator	Field coordination, route/clinic coordination, driver and clinic operational issues.
Regional Pharmacist	Approve vaccine-specific temperature rules and support excursion-related clinical decisions.
Medical Director	Approve incident-response protocol and clinical governance.
Legal Team	Data ownership, access, retention, governance and compliance issues.
Ministry Cold Chain Technical Advisor	Review alignment with ministry cold-chain requirements.
Drivers and Clinic Nurses	Identify operational problems, respond to alerts according to approved procedures, and provide user feedback.

9.1 Escalation Rules

1. A team member identifies a new risk, trigger, incident, or significant change in an existing risk.
2. The risk owner records the condition and assesses whether the existing score remains appropriate.
3. If the risk is Medium or above, the owner defines or updates a preventive action and target date.
4. High and Critical risks are reviewed during project-control meetings and tracked until the response is implemented.
5. Clinical, vaccine-rule, regulatory, budget, or ministry-related decisions are escalated to the appropriate authority.
6. If a risk becomes an actual event that affects the project, it is also recorded as an issue/incident and managed through the appropriate project process.
7. After the response, the residual risk is reassessed and the register is updated.

10. Risk Management by Project Phase

Phase	Relevant risks	Risk management activities
Weeks 1–2: Requirements & Approval	R09, R10, R11, R12, R14, R15	Confirm vaccine rules, incident response, data governance, certification requirements, budget and project governance; establish baseline risk register.
Weeks 3–5: Development & Configuration	R01, R03, R04, R05, R09, R13	Build offline functions, rule engine, records and reports; test technical controls; control scope.
Weeks 6–7: Testing & Pilot Preparation	R01, R02, R03, R04, R05, R07, R08	Run offline, excursion, battery, integration and UAT tests; confirm fallback procedures and training.
Weeks 8–11: Field Pilot	All high-priority risks, especially R01–R08 and R13	Monitor shipments, incidents, device failures, alerts, user feedback, and operational readiness; record deviations.
Week 12: Evaluation & Recommendation	R03, R04, R05, R07, R08, R10, R13	Review temperature records, incidents, device reliability, response actions and feedback; document residual risks and rollout considerations.

11. Risk Monitoring and Control

Risk management should continue throughout the project rather than being completed once at the planning stage.

11.1 Risk Review Frequency

Review point	Recommended activity
During requirements/approval work	Review risks whenever a requirement, vaccine rule, certification condition, budget assumption, or governance requirement changes.
Before each testing cycle	Confirm that technical and clinical risks have test cases and acceptance criteria.
Before each field delivery	Perform device readiness, battery, configuration and fallback checks.
Weekly during field pilot	Review incidents, alerts, device reliability, user feedback, pending sync records and new risks.
At pilot close	Assess which risks occurred, which controls worked, residual risks, and lessons for any wider rollout.

11.2 Suggested Risk Indicators

- Number of missed temperature readings or incomplete shipment records.
- Number of false alerts and missed excursions.
- Number of device failures and average recovery time.
- Number of deliveries affected by power/battery problems.
- Number of records awaiting synchronisation after connectivity returns.

- Number of unauthorised or unexpected vaccine-rule changes.
- Number of unresolved incidents and time taken to assign a response owner.
- User acceptance issues reported by drivers, nurses and depot staff.
- Number of scope-change requests and their estimated schedule impact.
- Status of certification, data governance, budget, and required approvals.

12. Contingency and Fallback Procedures

The pilot documentation already provides several fallback principles. These should be treated as part of the risk response design rather than as optional actions.

Event	Immediate response	Follow-up
Connectivity unavailable	Continue local monitoring and local alerts; store records locally.	Synchronise records when connectivity returns and verify that data is complete.
Device loses power	Follow the approved fallback procedure and use paper logging where necessary.	Record the event, check charging/battery controls, and assess whether shipment data is complete.
Device failure	Use the temporary paper backup and report the device failure as an incident.	Replace/repair device if appropriate; investigate failure and update reliability statistics.
Temperature excursion detected	Show/communicate the alert and follow the approved incident-response and escalation protocol.	Clinical/pharmacy authority determines the appropriate handling; record incident details.
Unclear vaccine rule	Do not rely on an unapproved rule for a clinical decision.	Escalate to the Regional Pharmacist/appropriate clinical authority and record the approved rule.
Major scope request	Pause acceptance of the request until scope impact is assessed.	Escalate to Project Sponsor if it affects budget, schedule, clinical scope, or pilot boundary.
Certification issue discovered	Do not proceed with deployment if required certification has not been confirmed.	Escalate to sponsor/technical reviewers and reassess device selection or schedule.

13. Residual Risk and Acceptance

Risk controls cannot eliminate every risk. After mitigation, the team should reassess the remaining risk. Residual risk is the level of risk that remains after planned controls are applied.

Risk	Expected residual condition after controls	Acceptance / escalation consideration
Connectivity	Internet may still be unavailable, but core monitoring/alerting should continue locally.	Acceptable for pilot if offline monitoring and later synchronisation are demonstrated.
Power	Clinic power may remain unreliable, but devices should begin journeys fully charged and have sufficient battery capacity.	Acceptable only if battery/readiness controls and fallback logging are available.
Device failure	A device can still fail despite testing and spares.	Residual risk should be reviewed using failure rate and impact on temperature records.
False alert	Some false alerts may remain possible.	Must be monitored; repeated or unsafe patterns require rule/test review.

Missed excursion	Testing reduces likelihood but cannot prove zero future failures in all conditions.	High clinical significance; any missed excursion should be escalated and investigated.
User acceptance	Some users may need additional training or workflow adjustment.	Review feedback and UAT outcomes before field approval.
Regulatory/data governance	Some requirements may remain pending until authorities confirm them.	Do not treat unresolved mandatory requirements as accepted without the relevant authority's decision.

14. Relationship Between Risk Management and Project Success

The risk plan is directly connected to the project's documented objectives and success criteria. The project aims to prevent undetected excursions from reaching patients, maintain complete digital temperature histories, provide local alerts without internet connectivity, and evaluate whether the system is practical for wider use.

Project objective / success criterion	Main risks affecting it	Key controls
Zero administered doses from a compromised shipment	R04, R05, R08, R09	Accurate alert logic, excursion testing, approved response protocol, approved vaccine rules, clear status/hold process.
Complete digital temperature history for every pilot shipment	R01, R02, R03	Offline storage, later synchronisation, battery readiness, device testing, temporary paper backup.
Clear local warning without internet/mobile connectivity	R01, R03, R04	Offline-first design, local alert hardware, excursion simulation, integration testing.
Evaluate practicality for wider use	R06, R07, R03, R04, R13	User testing, feedback collection, reliability metrics, controlled scope, pilot evaluation.

15. Risk Management and Change Control

Because the project uses an incremental development approach and the scope contains several functional modules, changes may occur as the team refines requirements. However, changes should not be accepted without considering their risk impact.

8. Describe the requested change and its purpose.
9. Identify affected requirements, deliverables, schedule tasks, budget items, and stakeholders.
10. Check whether the change introduces a new risk or increases an existing risk.
11. Assess whether the change affects patient safety, clinical governance, device certification, data governance, or pilot scope.
12. If the change is outside the approved pilot boundary, escalate it to the Project Sponsor.
13. Update the risk register and schedule if the change is approved.

This is particularly important for VaxiFlow because the Feature Scope lists functions such as authentication, user management, dashboards, shipment management, delivery-stop management, vaccine-box management, notification contacts, cold-box management, temperature monitoring/alerts, handover, shipment completion, and shipment history/reporting. The project should distinguish between functions required to demonstrate the pilot and functions that may be deferred if they threaten the core safety objectives.

16. Open Risk Items and Assumptions to Confirm

Open item	Why it matters to risk	Required confirmation
Final approved budget	Cost uncertainty can affect procurement, software work and scope.	Finance team / Project Sponsor.
Device certification requirements	An unsuitable or uncertified device could delay deployment.	Technical/compliance reviewers and relevant authority.
Exact vaccine tolerance rules	Incorrect rules can create unsafe or unnecessary decisions.	Regional Pharmacist and relevant clinical/pharmacy staff.
Data-retention requirements	Retention and ownership affect architecture and compliance.	Legal, medical and ministry stakeholders.
Formal Project Manager appointment	Clear authority is needed for day-to-day coordination and escalation.	Project Sponsor / project governance.
Final technical platform/vendor	The Business Case states these have not yet been selected.	Project/technical team after requirements and feasibility review.
Final incident-response protocol	Alert detection must be linked to a clear operational response.	Medical Director and relevant operational stakeholders.

17. Risk Register Update Template

Use the following fields when a new risk is identified or an existing risk changes:

Field	What to record
Risk ID	Unique identifier such as R16.
Date identified	Date the risk was first recorded.
Risk statement	Cause → uncertain event → potential effect.
Category	Technical, clinical, data, schedule, etc.
Likelihood / Impact	1–5 rating with brief justification.
Score / Priority	Likelihood × Impact and priority band.
Response strategy	Avoid, reduce/mitigate, transfer if applicable, accept, or contingency.
Actions	Specific preventive or contingency actions.
Owner	Person responsible for monitoring and response.
Trigger	Observable condition that indicates the risk is occurring or increasing.
Due date	Date by which the response should be completed.
Status	Open, monitoring, mitigated, occurred/issue, closed.
Residual risk	Remaining likelihood/impact after controls.

18. Conclusion

VaxiFlow's risk profile is shaped by its purpose, pilot environment, and technical design. The project must operate on a rural route where connectivity can be unavailable, two pilot clinics have unreliable power, temperature rules require clinical approval, device certification is still to be confirmed, and the pilot must demonstrate results within approximately three months. These conditions make proactive risk management essential.

The most important controls are therefore closely connected to the project design: offline monitoring and alerts, local data storage with later synchronisation, battery and device readiness checks, thorough excursion testing, approved vaccine-specific rules, a clear incident-response process, user involvement, temporary paper backup, and strict control of pilot scope. Risk management should remain active from requirements and approval through development, testing, field deployment, and final evaluation.

Final project-management principle

The goal of risk management is not to remove every uncertainty. The goal is to identify the uncertainties early, prepare practical responses, assign responsibility, and make sure that risks that could affect patient safety, pilot readiness, or core project objectives receive attention before they become failures.

Appendix A – Source Alignment

This plan was prepared from the project materials supplied for VaxiFlow. The main source-derived elements used are summarised below.

Project material	Used for this risk plan
Project Charter	Project purpose, objectives, success criteria, high-level requirements, scope boundaries, high-level risks, milestones, governance, stakeholders, approval requirements, constraints and assumptions.
Business Case	Problem context, alternative solution assumptions, scope/deliverables, cost assumptions, documented risk list, constraints, stakeholder roles and implementation phases.
Feature Scope	Functional areas including authentication/access control, user management, dashboard, shipment management, delivery stops, vaccine/cold-box management, temperature monitoring/alerts, handover, completion, and history/reporting.
Presentation Work	Development cycle selection, proposed team roles, WBS, task structure, technical modules, testing activities, deployment activities, and risk/schedule control role.
Gantt Chart	Planned sequencing and timing of project tasks from September 14, 2026 onward, including requirements, design, development, testing, deployment and evaluation activities.
Important Links	Project website, presentation and Figma mock-screen references as supporting project resources.